

The Prediction tab, in plain English

The app's first three tabs answer *what happened* and *what has been happening*. The Prediction tab answers **what is likely to happen next** — using classical statistics retrained on the live incident database. No AI calls: predictions are instant, free to run, and identical every time.

The whole tab in one breath

Imagine a jar of marbles for every month — one marble per incident. **Model 1** looks at the last 21 jars (trusting the newest ones most) and guesses how full the next jar will be — not one exact number, but "probably between here and here." **Model 2** is an experienced safety teacher who has watched all 647 incidents and can tell you, for any situation, how likely an incident there is to be the *bad* kind — as a number out of 100 instead of a gut feeling. **Model 3** is the same teacher remembering *why* things usually go wrong in that situation — but she has only had 245 incidents fully explained to her, so she calls it her best guess. Every new upload teaches all three of them; they never stop learning.

647

incidents on record
Aug 2023 – Apr 2025

3

prediction models
forecast · severity · root cause

0

AI calls at prediction time
pure statistics

"How many incidents next month?"

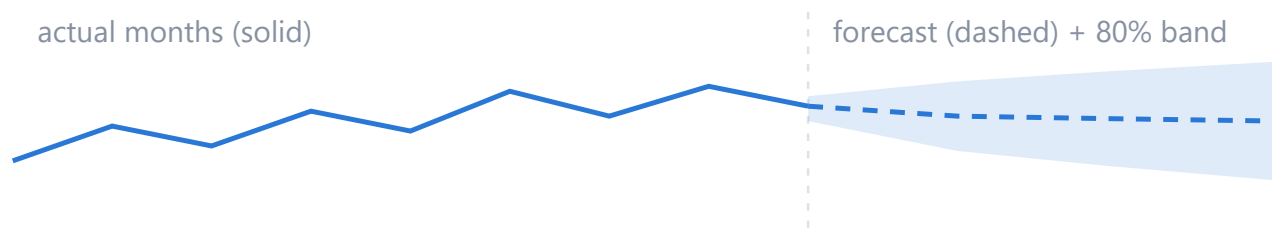
The model — **exponential smoothing (ETS)** — is a smart running average. Each month it blends the newest actual count with its previous estimate, so **recent months matter exponentially more than old ones**. It separately tracks whether the level is drifting up or down (the trend), and dampens that trend into the future so a rising line levels off rather than climbing forever.

$$\text{level}_t = \alpha \cdot \text{actual}_t + (1 - \alpha) \cdot (\text{level}_{t-1} + \phi \cdot \text{trend}_{t-1})$$

$$\text{trend}_t = \beta \cdot (\text{level}_t - \text{level}_{t-1}) + (1 - \beta) \cdot \phi \cdot \text{trend}_{t-1}$$

$$\text{forecast (h months out)} = \text{level} + (\phi + \phi^2 + \dots + \phi^h) \cdot \text{trend}$$

α, β — how fast old months fade (with $\alpha = 0.3$: last month counts 30%, the one before $\approx 21\%$, then $\approx 15\%$...). ϕ — the damping factor, just below 1, which flattens the trend over the horizon. All three are fitted automatically by finding the values that would have best predicted the 21 months we already have (maximum likelihood).



The band is the model saying "80% confident the real number lands in here." It widens with distance — further ahead is genuinely harder, and the band comes from how large the model's own past errors were.

Live example: next month it expects **≈ 34 incidents**, 80% confident the true count lands between **23 and 46**.

Built-in upgrade: yearly rhythms (winter weather, summer works) need two full years of history to be told apart from coincidence. At 24+ months of data the model switches on seasonality automatically and says so in the chart caption. Currently: 21 months.

"How bad would an incident in this situation be?"

Pick a scenario — route, shift, incident type, person involved, near-miss or not — and the model estimates the probability the incident turns out **worse than "Minor"**. It has learned, from all 647 incidents, which circumstances push risk above or below the 13% historical average — the same instinct an experienced safety manager has, but learned from data and expressed as a number.

score $z = w_1 \cdot x_1 + w_2 \cdot x_2 + \dots + b$ (the linear part)

risk $p = 1 / (1 + e^{-z})$ (the sigmoid squash)

Each **x** is a yes/no fact about the scenario (route = North West? shift = Night?...); each weight **w** is learned "risk points" for that fact. The S-shaped sigmoid maps any score into a valid probability between 0 and 1 — this is **logistic regression**, linear regression's sibling built for probabilities.

Live example: a night shift on the North West route involving an employee scores **36% risk vs the 13% average** — nearly three times baseline. Model quality: cross-validated AUC **0.68** (0.5 = coin toss, 1.0 = perfect) — real signal, not a crystal ball.

"If something goes wrong here, why will it probably go wrong?"

Same scenario, different question: a probability bar chart over root-cause categories — equipment, environment, procedures, fatigue... It is the same logistic-regression math extended to eight possible answers (**softmax**: one score per factor, converted into percentages that sum to 100%).

It can only learn from incidents whose root cause has already been categorized in the first tab — currently **245 of 647** — so the UI honestly labels it an *indicative profile*, not a verdict. **This is the flywheel:** every categorization run enlarges its classroom, and it retrains automatically after each upload.

"Is this linear regression?"

Close — two of the three models are its sibling. Plain linear regression predicts unbounded numbers: it would happily report a "130%" or "–12%" risk. **Logistic regression** keeps the linear weighted-sum core but squashes the result through the sigmoid so it is always a valid probability — that's models 2 and 3. Model 1 (the forecast) isn't regression at all: it is a **recursive weighted average** that discounts the past exponentially. So: one weighted-averaging model, two logistic regressions, zero black boxes.

PRESENTING IT

The five-step demo script

1. **Open on the headline numbers** — "647 incidents on record; here is next month's expected volume before it happens."
2. **Drag the horizon slider** — the widening band is the model being honest about uncertainty, a feature, not a flaw.
3. **Filter to one route** — show forecasts localize.
4. **Run a what-if the client recognizes** — their route, a night shift — and read off severity risk and likely root causes.
5. **Close on the flywheel** — every upload retrains every model; the product sharpens itself with use.

HONESTY BOX

Caveats to volunteer before they're asked

- **The models trust the uploads.** A month whose incidents were never uploaded reads as a genuinely quiet month.
- **Probabilities, not promises.** "36% risk" means roughly 1 in 3 similar historical incidents were serious — not that this one will be.
- **Small data, wide bands.** 647 incidents over 21 months is honest-but-uncertain territory; every new upload tightens the estimates. The factor model in particular is still learning — with 245 labeled examples it is indicative only.

If asked "why not AI/deep learning?" — neural networks need thousands of examples and would memorize noise at this scale. The right tool for 21 months and 647 rows is exactly what's used here; the LLM does what it is best at (reading investigation text) and statistics does what it is best at (forecasting and risk).

The three models at a glance

MODEL	QUESTION	METHOD	TRAINED ON	QUALITY TODAY
Volume forecast	How many next month?	ETS, damped trend (seasonality auto-enables at 24 months)	21 monthly counts	80% band $\approx \pm 11$ incidents at 1 month
Severity risk	How bad, in this context?	Logistic regression	All 647 incidents	AUC 0.68 (0.5 = chance)
Factor profile	Why, in this context?	Multinomial logistic (softmax)	245 categorized incidents	Indicative — improves with each categorization run

Prepared July 2026 · Figures (incident counts, model scores) reflect the database at time of writing; the in-app "Demo notes" panel always shows live values. Prediction stack: statsmodels (ETS) + scikit-learn (logistic regression) — no external services.